

Citation Faithfulness in Open-Weight Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) systems often use citations to show that an answer is grounded in retrieved documents. This can be misleading. A citation can be correct because the cited text agrees with the answer, but it can still be unfaithful if the model already knew the answer and added the citation afterward. We study this gap in three open-weight instruction-tuned models: Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Gemma-3-12B-it. Our experiments combine four parts: a Natural Questions and KILT Wikipedia benchmark for citation post-rationalization, a ConflictBank validation set, a Llama-only activation-patching study on PopQA capital-city questions, and linear probes trained on hidden states. Random forged documents are rarely cited, but forged documents that look more relevant are cited more often. Qwen has the highest post-rationalization rate in the cited-other condition (21.6%), followed by Gemma (17.8%) and Llama (9.1%). ConflictBank shows that all three models strongly follow supplied context, even when it conflicts with their stored knowledge. Linear probes predict the behavioral label better than simple lexical and semantic baselines, with AUROC values of 0.943 for Llama, 0.969 for Qwen, and 0.831 for Gemma. The mechanistic validation did not produce successful causal edits in this run. This suggests that the selected components were not enough to reliably control citation behavior during full generation. Overall, the results show that citation faithfulness is different from citation correctness. It can be measured through behavior and partly predicted from internal model states.

1 Introduction

Retrieval-augmented generation (RAG) combines language models with retrieved documents [9, 6, 7]. It matters because a model cannot store every useful fact in its weights. Its knowledge can also become outdated, and it can hallucinate answers that sound plausible but are not supported. Retrieval helps by giving the model external documents that may be more current, more specific, and easier to check than the model’s stored knowledge. In practical RAG systems, citations connect an answer to its claimed evidence. They let readers inspect the source of a claim, and they are often treated as signs that the cited material actually supported the answer. This is especially important in information-seeking settings, where users may rely on citations to decide whether an answer can be trusted [11].

A cited answer can look transparent while hiding several separate steps. First, the retriever selects documents. Then the generator writes an answer using both the documents and the knowledge stored in the model. Finally, the system adds a citation marker that points to one of the retrieved sources. Doing the last step well does not mean the earlier steps worked as intended. For example, a system may retrieve a useful passage, ignore it, answer from memory, and then attach a citation that happens to match the answer. To a reader, the output may look grounded even

though the citation does not explain how the answer was produced. This matters because citations are not just formatting. They are often used as a practical sign of provenance, auditability, and reliance on external evidence.

Producing citations is not the same as proving that the citations reflect what the model used. Attributed-generation systems have made strong progress on writing answers with citations and checking whether cited passages support individual claims [4]. These checks mainly measure citation correctness or completeness: whether a passage supports a claim, and whether important claims have support. They do not show whether the model used that passage instead of relying on stored knowledge or a shallow match between the document and the answer. A citation can therefore be correct in content but still be a poor explanation of the model’s evidence use.

We use *citation correctness* to mean the text relation between a claim and a cited passage. We use *citation faithfulness* to mean whether the citation matches the evidence the model used while generating the answer. This is a causal question, not just a writing-style question. Correctness can be checked after generation by comparing the answer and the passage. Faithfulness requires asking what changes when the evidence role of a document changes. No single test answers this perfectly. Behavioral interventions can show whether a model is sensitive to forged support. Activation patching can test specific internal mechanisms. Probes can show whether a risk signal is present in hidden states. These methods answer different questions, so we treat them as complementary.

Wallat et al. formalize this distinction for RAG attributions and introduce post-rationalization interventions [22]. In these interventions, an answer-bearing string is inserted into a document that was not the original factual source. If the model repeats the target statement and cites that document, the citation is hard to read as true provenance. This connects to work that studies attribution from inside the model, such as context-sensitive token analysis and saliency-based document attribution [17]. Together, these ideas motivate evaluations that separate textual support from actual reliance on retrieved context.

Open-weight instruction-tuned models are useful for this kind of study. They let us run controlled generation experiments across model families, and they also give access to hidden states and intermediate activations. This means we can compare three things: how models behave under document interventions, what information is present inside their representations, and whether selected internal components can change citation behavior. Still, access to activations does not make faithfulness directly visible. A good probe shows predictability. A good patch shows a causal effect under one intervention. Neither one alone proves that a citation fully explains the model’s reasoning.

We organize the study around three questions. First, how often does a model cite an adversarial document when that document contains the answer phrase but was not the original source of support? Second, does this behavior become more common when the forged document looks more relevant? Third, can we detect post-rationalization from hidden states, and can activation patching find components that reliably change citation behavior during full generation? We also use a knowledge-conflict experiment to test how strongly each model follows supplied context when it disagrees with stored knowledge.

This paper studies citation faithfulness in three open-weight instruction-tuned RAG models: Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Gemma-3-12B-it. We make four contributions. First, we measure post-rationalization on Natural Questions with KILT Wikipedia retrieval under random, relevant-uncited, and cited-other document interventions. Second, we use ConflictBank to study what models do when supplied context conflicts with stored knowledge. Third, we test a focused causal hypothesis about Llama-3.1-8B-Instruct with activation patching on PopQA capital-city examples. Fourth, we test whether post-rationalization labels can be decoded from hidden states, and whether these probes beat lexical, logit-margin, and embedding-similarity baselines.

Together, these experiments separate behavior, causal mechanisms, and prediction without treating any one of them as the same thing as citation correctness.

The rest of the paper is organized as follows. Section 2 reviews work on RAG, attributed generation, knowledge conflicts, mechanistic interpretability, and probing. Section 3 describes the research architecture, datasets, models, and experimental methods. Section 4 presents the behavioral, conflict, mechanistic, and probe results. Section 5 discusses what the results mean for citation faithfulness. Sections 6 and 7 describe limitations and conclusions.

2 Related Work

Early RAG systems joined language models with external retrieval for knowledge-intensive tasks. REALM adds retrieval during language-model pre-training. RAG generates while marginalizing over retrieved passages. Fusion-in-Decoder lets the decoder combine evidence from several passages [6, 9, 7]. These systems showed that retrieval can improve factual and open-domain question answering. However, the usual metrics mainly measure answer quality. Natural Questions provides real user questions with annotated answers, and KILT gives a shared Wikipedia source with provenance annotations [8, 16]. These datasets make retrieval and source tracking easier to evaluate. Still, a relevant passage and a correct answer do not prove that the generator relied on that passage for a specific claim.

Attributed generation makes sources visible by asking systems to attach references to generated claims. Gao et al. compare inline and post-hoc citation generation, and they evaluate citation quality through claim-level support and completeness [4]. Liu et al. study generative search engines from the reader’s point of view. They measure whether claims are supported by citations and whether users can verify those claims [11]. This work gives important standards for useful citations: important claims should be covered, cited sources should support them, and references should be easy to inspect. But these standards mainly compare the final claim with the final source. They do not test whether the source helped produce the claim. A model can answer from memory and then attach a passage that supports the answer. That citation may be correct, but it may still give a misleading account of provenance.

Citation faithfulness asks whether an emitted citation reflects the evidence that influenced generation. Wallat et al. define the difference between correctness and faithfulness in RAG and introduce adversarial interventions [22]. Their method places answer-bearing strings in documents that were not the original source of support. If a model repeats the target statement and cites that inserted document, this is behavioral evidence of post-rationalization. Qi et al. study faithful attribution from inside the model by using context sensitivity and saliency to estimate which retrieved documents affect answer tokens [17]. These approaches answer different questions. Adversarial tests show behavior under controlled input changes. Internal attribution methods estimate influence in one forward pass.

The difference between retrieved evidence and stored model knowledge also appears in work on knowledge conflicts. PopQA studies when models answer from stored knowledge and when outside evidence helps, with entity popularity as an important factor [12]. ConflictBank tests cases where supplied context conflicts with a model’s stored knowledge and measures which source the model follows [20]. Following the supplied context is related to citation faithfulness, but it is not the same thing. A context-matching answer shows that the model is sensitive to the supplied evidence. A faithful citation must also point to the document that actually supported the claim.

Mechanistic interpretability gives tools for testing causal ideas inside a model. Causal tracing and activation editing have found transformer components involved in factual recall, and they show

that changing internal states can change model predictions [13]. TransformerLens provides hooks and activation caches that are commonly used for these patching experiments [15]. The closest related work is by van Dort and Heuss, who study citation generation in Llama-3.1-8B-Instruct with activation patching on PopQA and a citation-token logit-difference metric [21]. Their work shows how activation patching can be used to study citation-token behavior in a controlled setting.

Linear probes ask a different question: can a simple classifier recover a target property from a model representation? Prior work has used such methods to study whether language-model activations contain information about truth and uncertainty. Burns et al. recover latent answers to true-or-false questions by finding directions in activation space that satisfy logical consistency constraints, even without labeled examples [3]. Azaria and Mitchell train classifiers on hidden activations to distinguish true from false statements, including statements generated by the model itself [1]. Li et al. train linear probes on attention-head activations to locate directions associated with truthful answers and then use those directions for inference-time intervention [10].

These studies show that internal representations can support prediction of factual truth, latent knowledge, or semantic uncertainty. Our probe target is different: it is not whether an answer is factually correct or whether the model is uncertain, but whether a citation is post-rationalized under a controlled document intervention. This places citation faithfulness within the broader use of hidden-state probes for model monitoring while introducing an attribution-specific behavioral label. As with probing more generally, strong classification performance shows that information linked to the label is linearly available; it does not prove that the model uses the probed direction or that the direction causes the behavior [2]. Probe evidence must therefore be interpreted separately from causal intervention evidence.

Building on these lines of work, this study uses established RAG datasets, citation-intervention methods, knowledge-conflict evaluation, activation patching, and linear probing to study citation faithfulness in open-weight models. The goal is to advance this area by connecting these methods in one experimental pipeline and using them to separate citation correctness from the stronger question of whether a citation reflects the evidence used during generation.

3 Methodology

3.1 Research Architecture

Figure 1 summarizes the experimental architecture. The main pipeline starts with Natural Questions, KILT Wikipedia, and ConflictBank. It retrieves top-5 BM25 contexts from 100-token Wikipedia chunks, runs deterministic generation with the three open-weight instruction-tuned models, applies behavioral interventions, and evaluates post-rationalization with citation labels and hidden-state probes. A separate auxiliary path uses PopQA capital-city pairs for Llama-only activation patching, component selection, and generation-level validation. This keeps the all-model behavioral comparison separate from the narrower Llama causal analysis.

3.2 Task Setup

Each example begins with a question and a document collection. Let q_i be the question for example i , and let $\mathcal{D} = \{d_1, \dots, d_N\}$ be the document corpus. The retriever ranks documents with a BM25 score $s(q_i, d)$ and returns the top- k documents:

$$R_k(q_i) = \text{TopK}_{d \in \mathcal{D}} s(q_i, d), \quad (1)$$

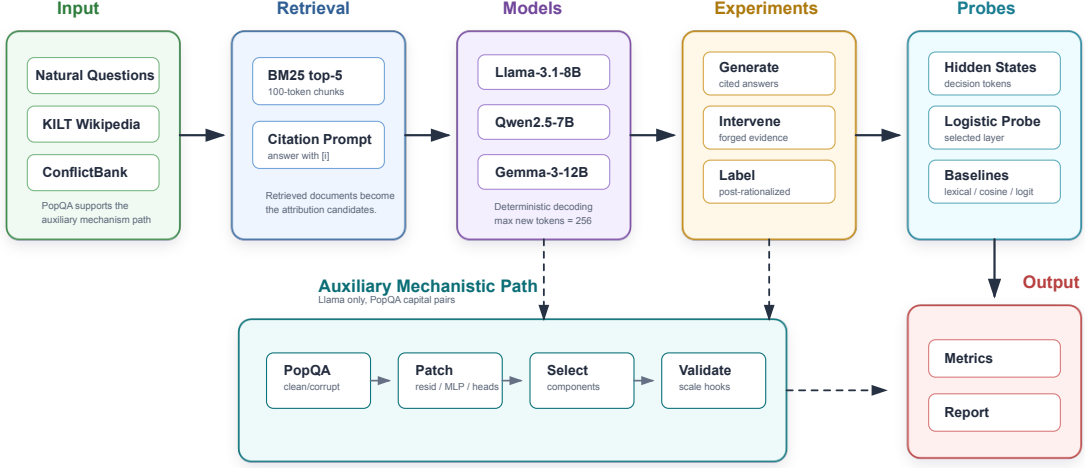


Figure 1: Research architecture for the citation-faithfulness experiments. The main path evaluates retrieval, generation, behavioral interventions, and probes across all three models. The auxiliary mechanistic path performs Llama-only activation patching on PopQA examples.

In this study, $k = 5$. We write the retrieved set as $C_i = \{d_{i1}, \dots, d_{ik}\}$. The model receives the question and these retrieved documents as a prompt $x_i = (q_i, C_i)$ and generates an answer:

$$y_i = M_\theta(x_i), \quad (2)$$

Here, M_θ is one of the instruction-tuned models. The generated answer contains one or more statements and citation markers. For a target statement t_i and a cited document index c_i , citation correctness can be written as a support relation:

$$\text{Correct}(t_i, c_i) = \mathbb{K}\{d_{ic_i} \models t_i\}, \quad (3)$$

where $d_{ic_i} \models t_i$ means that the cited document supports the statement. This only checks the final text. It does not show whether the model used that document while producing the answer.

To test faithfulness, we change the retrieved context and observe the model’s response. For each target statement, we create an intervention I_a that inserts an answer-bearing phrase into an adversarial document d_{ia} . The new retrieved context is

$$C_i^{(a)} = I_a(C_i), \quad (4)$$

and the model generates a new answer

$$y_i^{(a)} = M_\theta(q_i, C_i^{(a)}). \quad (5)$$

We then define two indicators. The first checks whether the target statement is recovered:

$$r_i^{(a)} = \mathbb{K}\{t_i \subset y_i^{(a)}\}. \quad (6)$$

The second checks whether the adversarial citation marker is used in the sentence that contains the recovered target phrase:

$$z_i^{(a)} = \mathbb{K}\{[a] \in \text{sent}(y_i^{(a)}, t_i)\}. \quad (7)$$

The post-rationalization label is therefore

$$\ell_i^{(a)} = r_i^{(a)} z_i^{(a)}. \quad (8)$$

A value of $\ell_i^{(a)} = 1$ means that the model recovered the target statement and cited the adversarial document. This is the main behavioral signal used in the study.

3.3 Models and Data

All model weights and tokenizers came from their official Hugging Face releases. We used three instruction-tuned checkpoints: `meta-llama/Llama-3.1-8B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`, and `google/gemma-3-12b-it`. These models were released by Meta, Alibaba’s Qwen team, and Google DeepMind, respectively [14, 18, 5]. Llama 3.1 and Gemma 3 require license acceptance. Qwen2.5 is distributed under the Apache 2.0 license. Because these are open-weight models, we can generate answers, extract hidden states, and run activation-patching experiments. For the semantic-similarity baseline, we used `sentence-transformers/all-mpnet-base-v2` to encode document-query pairs, following the sentence-transformer bi-encoder approach [19].

The behavioral question set contains 1,444 unique Natural Questions identifiers and questions released with the experiments of Wallat et al. [22]. The original questions come from Natural Questions [8]. Retrieved evidence came from the full KILT Wikipedia knowledge source [16]. We converted the Wikipedia pages into non-overlapping chunks of 100 whitespace tokens before BM25 retrieval. In short, Natural Questions supplied the questions, and KILT supplied the document corpus.

The mechanistic examples came from the `akariasai/PopQA` test split [12]. We kept only capital-city relations and built the clean and corrupted pairs described below. The counterfactual validation data came from the `Warrieryes/CB.qa` training split associated with ConflictBank [20]. We streamed that split, kept the first 2,000 examples with the expected schema, and sorted them by example identifier before generation. ConflictBank was used only for secondary validation. PopQA was used only for mechanistic analysis. Probe labels and features came from our own behavioral interventions, not from either dataset.

3.4 Models and Execution

We evaluated three open-weight instruction-tuned models: `meta-llama/Llama-3.1-8B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`, and `google/gemma-3-12b-it`. Behavioral and probe experiments were run for all three models. Mechanistic activation patching was run only for Llama-3.1-8B-Instruct, which matches the mechanistic scope of the study. Generation used greedy decoding with a maximum of 256 new tokens. The experiments ran on Apple MPS with backend-selected reduced precision and no quantization.

3.5 Behavioral Post-Rationalization Benchmark

The main behavioral benchmark uses 1,444 Natural Questions queries and top-5 BM25 retrieval from KILT Wikipedia. For each model, we first generated answers with citations. The system instruction required citations for document-supported claims. We then parsed the cited statements and built interventions around target statements that the model originally cited.

Each target statement was tested under three adversarial conditions:

- **Random**: the answer-bearing phrase is inserted into an otherwise random document, placed at citation index [5].
- **Relevant uncited**: the phrase is inserted into a retrieved document that is topically relevant but was not the original cited source.
- **Cited other**: the phrase is inserted into another document that the model cited elsewhere in the original answer.

An intervention is available when we can construct the adversarial document. We label the output as POST-RATIONALIZED when two things happen: the model recovers the target statement, and the adversarial document’s citation marker appears in the sentence with the recovered target phrase. The main metric is the adversarial citation rate, conditioned on statement recovery and condition availability:

$$\text{PRR} = \frac{\#\{\text{recovered target and adversarial citation}\}}{\#\{\text{condition available and recovered target}\}}. \quad (9)$$

Wilson 95% intervals were computed for the conditional rate.

3.6 ConflictBank Validation

We use ConflictBank as a secondary validation experiment, not as probe training data. ConflictBank examples contain a question, an original or parametric answer, a conflicting answer, and a piece of evidence that supports the conflicting answer. This lets us test what the model does when the retrieved context disagrees with what the model may already know.

For each example j , let q_j be the question, a_j^p be the original or parametric answer, a_j^c be the conflicting answer, and e_j^c be the conflicting evidence. Each model is run twice. First, it answers the question without any supporting document:

$$y_j^0 = M_\theta(q_j). \quad (10)$$

Second, it answers the same question with the conflicting evidence provided as the retrieved document:

$$y_j^c = M_\theta(q_j, e_j^c). \quad (11)$$

The generated answer is then assigned to one of three classes. After text normalization, it is labeled PARAMETRIC MATCH if it contains a_j^p but not a_j^c . It is labeled CONTEXT MATCH if it contains a_j^c but not a_j^p . It is labeled AMBIGUOUS if it contains both answers or neither answer:

$$g(y_j) = \begin{cases} \text{PARAMETRIC MATCH,} & a_j^p \subset y_j \text{ and } a_j^c \not\subset y_j, \\ \text{CONTEXT MATCH,} & a_j^c \subset y_j \text{ and } a_j^p \not\subset y_j, \\ \text{AMBIGUOUS,} & \text{otherwise.} \end{cases} \quad (12)$$

We apply this classifier to both y_j^0 and y_j^c . The main validation signal is the rate at which the with-context answer is labeled CONTEXT MATCH:

$$\text{CMR} = \frac{1}{n} \sum_{j=1}^n \mathbb{1}\{g(y_j^c) = \text{CONTEXT MATCH}\}. \quad (13)$$

In this study, $n = 2000$ for each model. A high context-match rate means that the model is strongly influenced by the supplied context when that context conflicts with stored knowledge. The resulting class counts are converted to percentages for the context-shift figure in Section 4.

3.7 Mechanistic Activation Patching

For mechanistic analysis, we built capital-city pairs from PopQA [12]. A clean prompt contains a document of the form “ A is the capital of S ” and asks the matching question. A corrupted prompt replaces the clean document with a distractor document. The distractor is length-matched in tokenization for the subject and answer spans. We selected the first 31 examples where the citation logit difference is positive in the clean prompt and negative in the corrupted prompt:

$$\Delta = \text{logit}(\text{" ["}) - \text{logit}(\text{" ."}). \quad (14)$$

For each selected example, we cached clean and corrupted activations. We then patched residual streams, MLP outputs, and attention heads. Denoising recovery and noising degradation were computed as

$$R_{\text{denoise}} = \frac{\Delta_{\text{patched}} - \Delta_{\text{corrupt}}}{\Delta_{\text{clean}} - \Delta_{\text{corrupt}}}, \quad D_{\text{noise}} = \frac{\Delta_{\text{clean}} - \Delta_{\text{patched}}}{\Delta_{\text{clean}} - \Delta_{\text{corrupt}}}. \quad (15)$$

We marked components as positive if their mean denoising recovery was at least 0.10. We marked components as necessary if their mean noising degradation was at least 0.20. We then tested the selected components by scaling them during generation on missed-citation and spurious-citation examples.

3.8 Linear Faithfulness Probes

For each model, we extracted hidden-state features from behavioral intervention examples labeled POST-RATIONALIZED or NOT-POST-RATIONALIZED. When an adversarial citation marker appeared, we used its first token as the decision position. Otherwise, we used the sentence position linked to the target phrase. For each layer, the feature vector was the mean hidden state over the decision token and the two answer tokens before it.

For layer λ and example i , let $h_{i,\lambda,t}$ be the hidden state at answer-token position t . If T_i is the decision token position, the probe feature is

$$u_{i,\lambda} = \frac{1}{3} \sum_{t=T_i-2}^{T_i} h_{i,\lambda,t}. \quad (16)$$

For each layer, the probe predicts the probability of post-rationalization with logistic regression:

$$\hat{p}_{i,\lambda} = \sigma(w_\lambda^\top u_{i,\lambda} + b_\lambda), \quad (17)$$

where σ is the sigmoid function. The probe is trained with balanced binary cross-entropy,

$$\mathcal{L}_\lambda = - \sum_i \alpha_{\ell_i} [\ell_i \log \hat{p}_{i,\lambda} + (1 - \ell_i) \log(1 - \hat{p}_{i,\lambda})], \quad (18)$$

where ℓ_i is the post-rationalization label and α_{ℓ_i} is the class weight. This weighting is used because post-rationalized examples are much rarer than non-post-rationalized examples.

We split questions by `question_id` into 70% training, 15% validation, and 15% test sets. This avoids leakage across interventions for the same question. We trained one L2 logistic regression

Model	Condition	Post-rationalized	Available	Rate	Wilson 95% CI
Llama-3.1-8B	Random	0	274	0.0%	[0.0, 1.7]
Llama-3.1-8B	Relevant uncited	8	245	3.5%	[1.8, 6.7]
Llama-3.1-8B	Cited other	2	23	9.1%	[2.5, 27.8]
Qwen2.5-7B	Random	3	326	1.1%	[0.4, 3.2]
Qwen2.5-7B	Relevant uncited	27	299	9.7%	[6.7, 13.7]
Qwen2.5-7B	Cited other	11	59	21.6%	[12.5, 34.6]
Gemma-3-12B	Random	4	829	0.6%	[0.2, 1.5]
Gemma-3-12B	Relevant uncited	13	726	1.9%	[1.1, 3.2]
Gemma-3-12B	Cited other	18	117	17.8%	[11.6, 26.4]

Table 1: Behavioral post-rationalization rates conditional on condition availability and target statement recovery.

probe per layer with balanced class weights. We selected the layer with the best validation AUROC and reported test AUROC, AUPRC, balanced accuracy, F1, and Brier score. The baselines used lexical overlap, citation-logit margin, and document-query cosine similarity from a sentence-transformer encoder.

4 Experimental Results

4.1 Behavioral Post-Rationalization

Table 1 reports conditional post-rationalization rates. Random forged documents are rarely cited. Rates are higher for relevant-uncited and cited-other documents. This suggests that models are more likely to cite forged evidence when it looks topically relevant or already fits the citation context.

The pattern is similar across the three model families. The random condition acts as a low-rate control. The cited-other condition is the strongest test because the document is already plausible in the model’s citation context. Qwen and Gemma are more vulnerable in this condition. Llama is more conservative, but the cited-other condition has low availability for Llama, so its confidence interval is wide.

4.2 ConflictBank

Figure 2 summarizes the ConflictBank outcomes. Each model has two bars: one for answers generated without supporting context, and one for answers generated after the conflicting document is supplied. The stacked colors show whether the answer matches the model’s parametric answer, matches the supplied conflicting context, or remains ambiguous under the string-matching rule. We use this figure instead of a separate ConflictBank table because the main result is the change between the two settings. Without context, most outputs are ambiguous because the short matching rule often finds neither answer directly. With conflicting evidence, all three models shift strongly toward the context-matching class. This shows that the supplied document has a strong effect on the final answer when it conflicts with stored model knowledge.

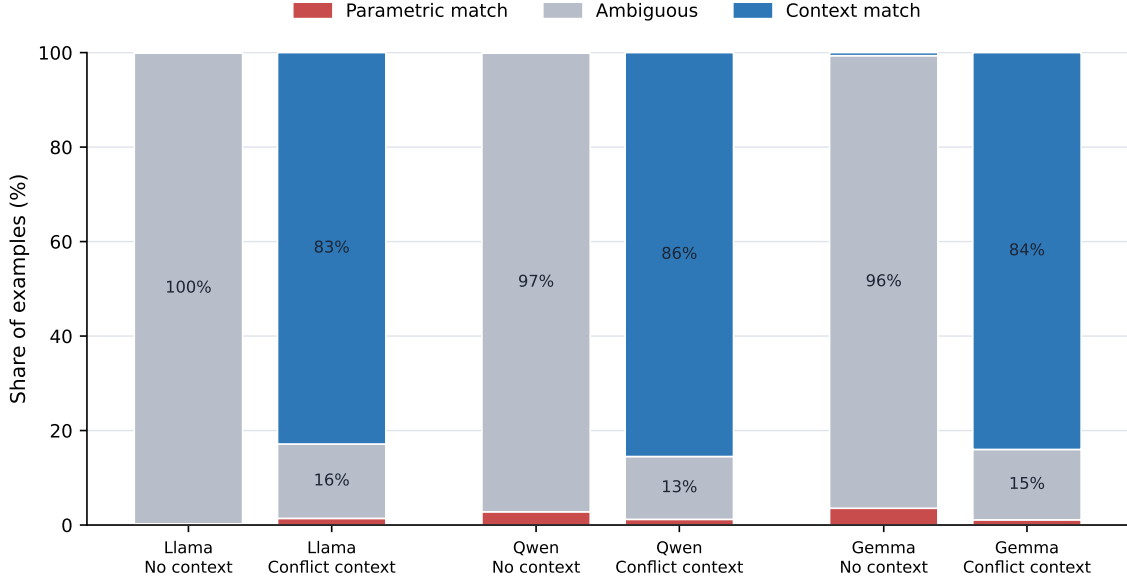


Figure 2: ConflictBank answer classes before and after conflicting evidence is supplied. The figure shows that all three models shift strongly toward the context-supported answer when conflict evidence is included.

Validation set	Count	Correction success rate
Missed citation	7	0.0%
Spurious citation	58	0.0%

Table 2: Mechanistic validation after component selection and scaling. Correction success rate is the percentage of examples whose citation behavior is fixed by the scaling intervention.

4.3 Mechanistic Patching and Interventions

The PopQA mechanistic phase selected 31 clean/corrupted pairs that met the filtering rule. Patching produced 512 residual entries, 512 MLP entries, and 1,024 attention-head entries. Component selection found no positive citation heads or MLP components under the denoising threshold. It found one necessary attention head, layer 0 head 3, and 11 necessary MLP positions in layer 0.

Validation found 7 missed-citation examples and 58 spurious-citation examples. We define correction success rate as the percentage of validation examples whose citation behavior is fixed after component scaling. For missed-citation examples, success means that the intervention adds the expected citation while preserving the answer. For spurious-citation examples, success means that the intervention removes the unwanted citation while preserving the answer. Component scaling did not fix either group: both correction success rates were 0.0. This negative result should be read carefully. Activation patching showed that early components affect the citation logit difference, but simply scaling the selected components was not enough to control full autoregressive generation.

4.4 Faithfulness Probes

The probe datasets contained 15,328 Llama feature rows, 16,912 Qwen feature rows, and 70,272 Gemma feature rows. Table 3 reports test performance for the selected layer of each model. The

Model	Layer	Rel. depth	AUROC	AUPRC	Bal. acc.	Best baseline AUROC
Llama-3.1-8B	21	0.68	0.943	0.200	0.479	0.757
Qwen2.5-7B	17	0.63	0.969	0.325	0.918	0.908
Gemma-3-12B	28	0.60	0.831	0.357	0.616	0.716

Table 3: Test performance of selected-layer linear probes. Best baseline AUROC is the best score among lexical overlap, citation-logit margin, and document-query cosine similarity.

selected layers fall around 60–68% relative depth. This suggests that the faithfulness signal is easiest to decode in middle-to-late residual streams.

The probes beat the strongest simple baseline by AUROC for all three models. Qwen is the clearest case, with high ranking performance and high balanced accuracy. Llama has high AUROC but low F1 and balanced accuracy at the default 0.5 threshold, mainly because it has very few positive test examples. Gemma has lower AUROC than Llama and Qwen but the highest AUPRC among the three, again under strong label imbalance.

5 Discussion

The behavioral results show that citation faithfulness is not simply present or absent. The models do not blindly cite arbitrary forged documents. However, they are more likely to cite adversarial documents when those documents are topically relevant or already plausible in the citation context. This suggests that citation selection can respond to document-answer compatibility even when the document is not the true source of the answer.

The ConflictBank results point in the same direction from another angle. When supplied evidence conflicts with stored model knowledge, all three models usually follow the supplied context. This is useful when the context is correct. It also means that document placement and surface evidence can strongly steer answers.

The mechanistic results are more mixed. Activation patching identifies early necessary components, but the later scaling intervention does not successfully control behavior. One possible reason is that citation decisions are distributed across many parts of the model and depend on context. Components that affect a next-token citation logit in a controlled prompt may not be enough to control a full multi-token answer. Another possibility is that local MPS precision and the simple scaling method made the validation too weak. Either way, the negative validation limits how strong the mechanistic claim can be.

The probe results are practically useful. Even with simple logistic regression, hidden states contain a signal that predicts post-rationalization. This does not prove that the probed feature causes unfaithful citation behavior. It does suggest that faithfulness risk could be monitored without rerunning the full adversarial intervention for every generation.

6 Limitations

This study uses one specific definition of citation faithfulness. It detects post-rationalization under answer-string insertion interventions. It does not cover every possible kind of unfaithful attribution. The Natural Questions setup depends on BM25 retrieval over KILT Wikipedia chunks and on parser rules for cited statements. The cited-other condition has low availability for Llama, so its confidence interval is wide. The mechanistic experiment is also narrow: it uses only Llama, only

PopQA capital facts, and a citation-token-versus-period logit difference as the target. The probe labels are imbalanced, especially for Llama and Gemma, so AUROC is more stable than F1. Finally, the local run used Apple MPS rather than CUDA, so exact numerical results may differ from the original CUDA-oriented setup.

7 Conclusion

This study investigated citation faithfulness in open-weight retrieval-augmented generation by distinguishing textual support from actual evidence use. A citation is correct when its document supports the generated claim, but it is faithful only when it reflects the evidence that influenced generation. We examined this distinction across multiple instruction-tuned model families through behavioral document interventions, knowledge-conflict evaluation, mechanistic activation patching, and hidden-state probing. The behavioral experiments showed that models rarely cited random forged documents, but cited them more often when they appeared relevant or plausible. The knowledge-conflict evaluation further showed that retrieved context can strongly influence outputs when it disagrees with stored knowledge. Together, these findings demonstrate that models can rely on supplied evidence while also attaching plausible citations that do not necessarily identify the source responsible for a claim. Document-answer compatibility, contextual influence, and faithful provenance are therefore related but distinct properties.

The internal analyses clarified both what can be detected and what can be controlled. Activation patching showed that an early attention head and several early MLP positions affected the model’s score for generating a citation token. However, scaling these components during full generation did not add missing citations or remove unwanted citations while preserving the answer. This means that changing a next-token citation score is not enough to control citation behavior across a complete generated response. Linear probes, by contrast, distinguished post-rationalized from non-post-rationalized outputs better than simple surface-level baselines, indicating that hidden states contain information useful for monitoring citation-faithfulness risk; however, this establishes predictability rather than causality and requires calibration before deployment. Taken together, the experiments show that citation faithfulness cannot be assessed through a single metric: support-based evaluation checks whether citations justify claims, controlled interventions test evidence reliance, conflict tests characterize sensitivity to context, mechanistic methods examine causal structure, and probes assess whether risk is internally detectable. Trustworthy RAG systems should therefore connect claims not only to documents that support them but also to evidence that genuinely shaped their generation.

Future work should test whether these findings generalize across retrieval methods, knowledge domains, model sizes, model families, and citation formats. Mechanistic studies should use more precise, position-aware interventions that target the selected activations without changing unrelated parts of the answer, and should evaluate whether effects remain stable across prompts and longer generations. Probe-based monitoring should be tested on independent datasets, calibrated under realistic class imbalance, and evaluated at decision thresholds suited to deployment rather than only by ranking metrics. Combining stronger causal tests with calibrated monitoring could support systems that detect unfaithful citations during generation and correct them without changing the factual content of the answer.

References

- [1] Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976, 2023.
- [2] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- [3] Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *The Eleventh International Conference on Learning Representations*, 2023.
- [4] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, 2023.
- [5] Google DeepMind. Gemma 3 model card. Official model documentation, 2025.
- [6] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*, pages 3929–3938, 2020.
- [7] Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 874–880, 2021.
- [8] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- [9] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, 2020.
- [10] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [11] Nelson F. Liu, Tianyi Zhang, and Percy Liang. Evaluating verifiability in generative search engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 7001–7025, 2023.
- [12] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 9802–9822, 2023.
- [13] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

- [14] Meta AI. Llama 3.1 model card. Official model documentation, 2024.
- [15] Neel Nanda and Joseph Bloom. TransformerLens. Software library for mechanistic interpretability, 2022.
- [16] Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, Majid Yazdani, Nicola De Cao, James Thorne, Yacine Jernite, Vladimir Karpukhin, Jean Maillard, Vassilis Plachouras, Tim Rocktaschel, and Sebastian Riedel. KILT: A benchmark for knowledge intensive language tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 2523–2544, 2021.
- [17] Jirui Qi, Gabriele Sarti, Raquel Fernández, and Arianna Bisazza. Model internals-based answer attribution for trustworthy retrieval-augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6037–6053, 2024.
- [18] Qwen Team. Qwen2.5-7B-Instruct model card. Official model documentation, 2024.
- [19] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992, 2019.
- [20] Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. ConflictBank: A benchmark for evaluating the influence of knowledge conflicts in large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [21] Ian van Dort and Maria Heuss. How do LLMs cite? a mechanistic interpretation of attribution in retrieval-augmented generation. In *Proceedings of the 48th European Conference on Information Retrieval*, 2026. To appear.
- [22] Jonas Wallat, Maria Heuss, Maarten de Rijke, and Avishek Anand. Correctness is not faithfulness in retrieval augmented generation attributions. In *ICTIR 2025: Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval*, pages 22–32. Association for Computing Machinery, 2025.